Sec. 26

Brief intro to infrared divergence.

We just got

$$\sigma = \sigma_{tree}\left[1 - \frac{11}{12}\alpha\left(\ln(s/m^2) + O((m^2)^{\frac{3}{2}})\right) + O(\alpha^2)\right].$$

no large logs.

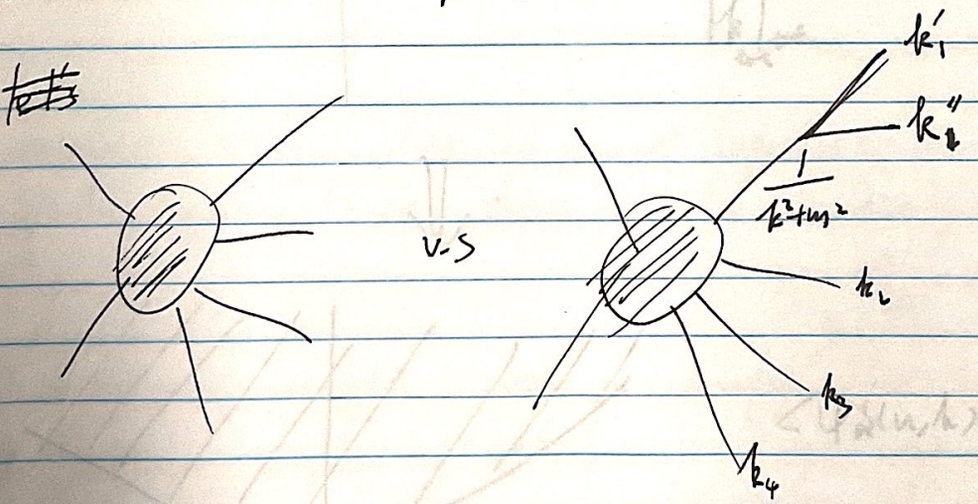
What if we are dealing with a theory with massless particles?

$m^2 = 0$.

The matrix amplitude would blow up., we encounter infinity again.

But this time, it is from IR. (as if it is there no matter how small s is, or if it is in the limit of $m \to 0$).

The cause are two folders.



If k_1' and k_2' are colinear or if k_2 is soft.,

the $\frac{1}{k^2+m^2} \to \frac{1}{k^2}$ can allow such configuration and blow-up.

For real detectable physical process, minimal Energy & separation angle are needed to separate.

identical particles.

So detectability regulates the process.

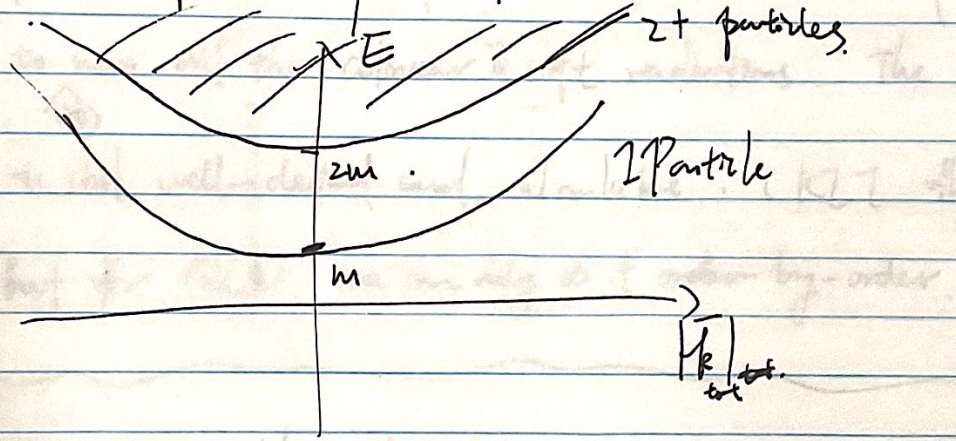
(very cheap way to discuss this profound topic)

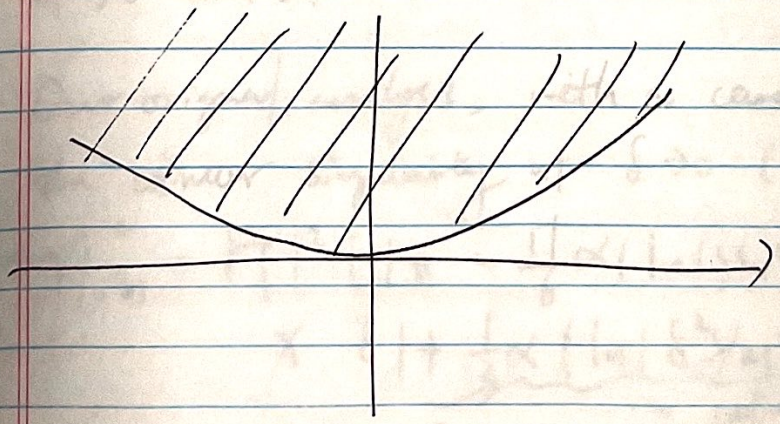
yet another.

But slightly more mathematically.

massless particles make one thing hard:

the sp separation of one particle & multiparticle state.





$\langle \varphi_{\vec{k}}(u,k) | \varphi(k) | 0 \rangle \neq 0.$

Many other trouble occurs, the singular part & $\rho(s)$ part of the exact propagator hard to separate.

1 particle state can funnel to multiparticle state., etc.

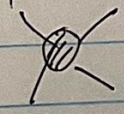
In real world, such situation happens.

An electron can emit infinite # of soft massless photons and still hit the electron pole (and it is a stable particle). One needs to redefine the state of electron to sum all the colinear & soft radiations. The EM theory (the) is still well-defined and calculable. (KLT theorem). [1]

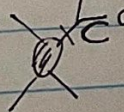
But for QCD, one can only do it order-by-order.

However, still, when we cannot separate two particles, in the massless limit, there is still a divergence in the $m^2 \to 0$ limit. $m^2/|k|^2 \ll \delta^2 \ll 1$.

For instance



σ_T



σ_{obs}

$\frac{g^2}{(k^2+m^2)^2}\frac{(2\pi)^{d-1}2\omega}{\delta^{d-1}(k-k_1-k_2)}\frac{1}{2}d\vec{k}_1 d\vec{k}_2$

Our original analysis, with a careful analysis to separate the colinear singularity at $\delta \to 0$ (δ open separation angle)

$$|T|^2_{obs} = |T|^2 \left[1 + \frac{1}{6}\alpha\left(\ln(s/m^2) + O(m^0)\right) + O(\alpha^2)\right]$$

$$\times \left[1 + \frac{1}{3}\alpha\left(\ln(\delta^2 s/m^2) + O(m^0) + O(\alpha^2)\right]\right.$$

$\sim$ colinear part from $n \to n+1$ process.

$$= |T_0|^2 \left[1 - \alpha\left(\frac{3}{2}\ln(s/m^2) + \frac{1}{3}\ln(1/\delta^2) + O(m^0)\right) + O(\alpha^2)\right]$$

Renormalization Schemes & RG flow.

We still didn't solve the case when we cannot resolve the particles, or in other words, we still have the $\ln(S/m^2)$ problem, (this time, with a different coefficient). What did we do wrong?

We need to review our renormalization procedure;

We had:

$$\Pi(k^2) = -\left[A+\frac{1}{6}\alpha\left(\frac{1}{\epsilon}+\frac{1}{2}\right)\right]k^2 - \left[B+\alpha\left(\frac{1}{\epsilon}+\frac{1}{2}\right)\right]m^2 + \frac{1}{2}\alpha\int_0^1 dx\, D\ln(D/\mu^2) + O(\alpha^2)$$

and

$$\Pi'(k^2) = -\left[A+\frac{1}{6}\alpha\left(\frac{1}{\epsilon}+\frac{1}{2}\right)\right] + \frac{1}{2}\alpha\int_0^1 dx\,[\ln(D/\mu^2)+1]x(1-x) + O(\alpha^2).$$

Previously we set ~~$\Pi(k^2)$~~ $\Pi(-m^2)=0$ and $\Pi'(-m^2)=0$ to keep the one-particle state normalized properly.

But in the $m^2\to 0$ limit, $D = x(1-x)k^2$

$\Pi(-m^2)=0$ automatically $\boxed{0\ln(0)=0}$

$\Pi'(-m^2)$ is $\ln(D/\mu^2)\to$ blows up. (ill defined)

as the one-particle pole overlaps with the multi-particle branch, so the pole of the exact propagator is no longer a simple pole.

The (only) way out is to change renormalization scheme.
[I hope you realize that ever after introducing path integral, the rest is just how to evaluate things in physics. (physically). This requires some consistent modifications in the Lagrangian density.]

Instead of setting A & B satisfying the b.c. $\Pi(-m^2) = 0$, $\Pi'(-m^2) = 0$, we set A and B just to cancel the divergencies.

$$A = -\frac{1}{6}\alpha\frac{1}{\epsilon} + O(\alpha^2)$$

$$B = -\frac{1}{6}\alpha\frac{1}{\epsilon} + O(\alpha^2)$$

This is called $\overline{MS}$ scheme., one of the most commonly used scheme.

Em-Es-bar

Modified (as we set $\mu = \sqrt{\mu}\sqrt{4\pi}\,e^{-\gamma/2}$)
Minimal Subtraction
Scheme.

So and our previous one was actually the On-Shell scheme.

So

$$\Pi_{\overline{MS}}(k^2) = -\frac{1}{12}\alpha(k^2 + 6m^2) + \frac{1}{2}\alpha\int_0^1 dx\, D\ln(D/\mu^2) + O(\alpha^2)$$

$$\Pi_{OS}(k^2) = -\frac{1}{12}\alpha(k^2 + m^2) + \frac{1}{2}\alpha\int_0^1 dx\, D\ln(D/D_0) + O(\alpha^2)$$

1 Not a new theory: $\Pi_{\overline{\text{MS}}} - \Pi_{\text{OS}} = a + bk^2$, a finite shift of A, B. What moves is the reference point: on-shell fixes m, residue 1 at $k^2 = -m^2$ and g at zero momentum, branch points once $m = 0$ (p. 4); $\overline{\text{MS}}$ uses an arbitrary μ. The legs' $\ln m$ then cancels against emission (p. 10); $\ln(s/\mu^2)$ is left for the RG.

(14)

$$D_0 = D\big|_{k^2=-m^2} = [x(1-x)k^2 + m^2]\big|_{k^2=-m^2}$$
$$= [-x(1-x) + 1]\, m^2.$$

Here $\Pi_{\overline{MS}}(k^2)$ has a well-defined $m \to 0$ limit. (1)

but $\Pi_{\overline{MS}}(-m^2) \neq 0$, $\Pi'_{\overline{MS}}(-m^2_{ph})$ will be evaluated separately.

~~so here.~~

Note. The physical quantity in the LSZ, scattering (measurable)

is the pole mass : m_{ph}.

And LSZ requires proper normalization, here φ is to be rescaled by $R^{-1/2}$ (if the residue is R) to keep LSZ normalized.

Let's see the consequence.

$$\Delta_{\overline{MS}}(k^2) = \frac{1}{k^2 + m^2 - \Pi_{\overline{MS}}(k^2)}$$

and by definition $k^2 = -m^2_{ph} \Longrightarrow$

$$-m^2_{ph} + m^2 - \Pi_{\overline{MS}}(-m^2_{ph}) = 0.$$

One can try to solve the integral differential equation, (typically very very hard), but we can do perturbation.

because $\Pi(k^2)$ is an integral differential because perturbation are around $-m^2$ not $-m^2_{ph}$.

Since we think/expect

$$m_{ph}^2 = m^2 - \Pi_{\overline{MS}}(-m_{ph}^2) + O(\alpha^2)$$

$\downarrow$

α order. so $m_{ph}^2 = m^2 + O(\alpha)$

$$= m^2 - \Pi_{\overline{MS}}(-m^2) + O(\alpha^2)$$

now we have

$$m_{ph}^2 = m^2 - \frac{1}{2}\alpha\left[\frac{1}{6}m^2 - m^2 + \int_0^1 dx\, D_0 \ln(D_0/\mu^2)\right] + O(\alpha^2)$$

$$= m^2\left[1 + \frac{5}{12}\alpha\left(\ln(\mu^2/m^2) + c'\right) + O(\alpha^2)\right]$$

↑ still have μ ↑ $c' = (134 - 3\pi\sqrt{3})/15 = 1.18$

(unphysical μ dependence & m.)

But μ is not a physical parameter.

How can a physical observable depends on ~~sup~~ unphysical quantities?

Something else is consistently varying m, α is!

We then know how α varies with μ.

$$\ln m_{ph} = \ln m + \underbrace{\frac{5}{12}\alpha\left(\ln(\mu/m) + \frac{1}{2}c'\right)}_{\text{here we view } (1+\delta) = e^{\delta}} + O(\alpha^2)$$

$$\ln(1+x) = x + \cdots$$

here we would assume $\frac{d\alpha}{d\ln\mu} = O(\alpha^4)$ for next page

1 Why may m, α run? μ only says how $\mathcal{L}$ is split into renormalized terms and counterterms. The bare $m_0 = (Z_m/Z_\varphi)^{1/2} m$ and g_0 (p. 13) are μ-free and the $\overline{\text{MS}}$ Z's depend on α only, so $\gamma_m = \frac{1}{2} d\ln(Z_\varphi/Z_m)/d\ln\mu = -\frac{5}{12}\alpha$, as below. $d\alpha/d\ln\mu = O(\alpha^2)$: α runs only through loops (p. 15).

A leap in our understanding is that unphysical parameters could depend on μ so $m(\mu)$, $\alpha(\mu)$ (1)

To keep physical parameter unchanged, by observing the form of the dependance, we require (as $\ln\mu$ and $\ln m$ contains infinite powers ~~negative~~)

$$\frac{d\ln M_{ph}}{d\ln\mu} = 0$$

$$= \frac{1}{m}\frac{dm}{d\ln\mu} + \frac{5}{12}\alpha + O(\alpha^2)$$

Where $\alpha \frac{d\alpha}{d\ln\mu}$ is $O(\alpha^2)$

$$\longrightarrow \frac{d\ln m}{d\ln\mu} = -\frac{5}{12}\alpha + O(\alpha^2)$$

called the anomalous dimension of mass parameter m, often dubbed $\gamma_m(\alpha)$.

$$\frac{dm}{d\ln\mu} = \left(-\frac{5}{12}\alpha + O(\alpha^2)\right) m.$$

If we try, e.g., $dM_{ph}/d\ln\mu$ or $dM_{ph}/d\mu$, there won't be clean expressions to get rid of the $\ln\mu$ dependance

Now get the quantity R. (residue)

$$R^{-1} = \frac{d}{dk^2}\left[\Delta_{\overline{MS}}(k^2)^{-1}\right]\Big|_{k^2=-M_{ph}^2}$$

$$= 1 - \Pi'_{\overline{MS}}(-M_{ph}^2)$$

$$= 1 - \Pi'_{\overline{MS}}(-m^2) + O(\alpha^2)$$

$$= 1 + \frac{1}{12}\alpha\left(\ln \mu^2/m^2 + \underbrace{C''}_{\uparrow}\right) + O(\alpha^2)$$

$(17-3\pi+3)/3 = 0.23$

$$-\frac{1}{12}k^2\alpha + \frac{1}{12}\alpha + \frac{1}{12}\alpha \ln \mu^2/m^2 \quad \text{by } \Pi'(k^2)\big|_{k^2=-m^2}$$

Also the vertex function using $\overline{MS}$.

$$C = -\alpha\frac{1}{\epsilon} + O(\alpha^2)$$

$$\rightarrow \quad V_{3\,\overline{MS}}(k_1, k_2, k_3) = g\left[1 - \frac{1}{2}\alpha \int dF_3 \ln(D/\mu^2) + O(\alpha^2)\right]$$

$$D_3 = \cancel{x+x_2 k_1^2}\; xy k_1^2 + yz k_2^2 + zx k_3^2 + m^2$$

Now we can assemble everything.

In $\overline{\text{MS}}$ scheme.

$$\mathcal{T} = R^2 \mathcal{T}_0 \left[1 - \frac{11}{12}\alpha\left(\ln s/\mu^2 + \mathcal{O}(m^0)\right) + \mathcal{O}(\alpha^2)\right]$$

R: normalization; $\mathcal{T}_0$: tree amplitude $\mathcal{T}_0 = -g^2\left(\frac{1}{s} + \frac{1}{t} + \frac{1}{u}\right)$

Plug in R, we get

$$\mathcal{T} = \mathcal{T}_0 \left[1 - \alpha\left(\frac{11}{12}\ln\frac{s}{\mu^2} + \frac{1}{6}\ln\frac{\mu^2}{m^2} + \mathcal{O}(m^0)\right) + \mathcal{O}(\alpha^2)\right]$$

$$|\mathcal{T}_{obs}| = |\mathcal{T}|^2\left(1 + \frac{1}{3}\alpha\left(\ln\delta^2 s/m^2\right) + \mathcal{O}(m^0) + \mathcal{O}(\alpha^2)\right]$$

$$= |\mathcal{T}_0|^2\left[1 - \alpha\left(\frac{3}{2}\ln\frac{s}{\mu^2} + \frac{1}{3}\ln\frac{1}{\delta^2} + \mathcal{O}(m^0)\right) + \mathcal{O}(\alpha^2)\right]$$

↓ ~~not~~ **no $\ln m$ dependence!**

a good theory in the $m \to 0$ limit.

$$-\left(\frac{1}{6}\ln\frac{\mu^2}{m^2}\right)\times 2 + \frac{1}{3}\ln\frac{s}{m^2} = 0\ln m^2 + c\ln\frac{s}{\mu^2}$$ [1]

but the observable cross section $|\mathcal{T}_{obs}|^2$ still depends on fake parameter μ., it can only be physical if we also get rid of it $\ln\mu$ dependence.

[1] Where p. 3's $\frac{1}{3}$ comes from: its margin integral gives, per line of energy $E = \sqrt{s}/2$ split at angle $< \delta$, $\frac{\alpha}{2}\int_0^1 dz\, z(1-z)\ln\frac{E^2\delta^2}{m^2} = \frac{\alpha}{12}\ln\frac{\delta^2 s}{m^2} + O(m^0)$; four lines, $\frac{\alpha}{3}$. Each cancels the $\ln m^2$ of one leg's R (p. 9). Soft emission is finite in $d = 6$: emission factor $\propto \omega^{-2}$, phase space $\omega^3 d\omega$.

$\alpha(\mu)$ is varying w.r.t. some good ~~rules~~ relations.
(again observing the structures & attempt to find simple relations)

$$\ln|\mathcal{T}|^2_{\text{obs}} = C_1 + 2\ln\alpha + 3\alpha(\ln\mu + C_2) + O(\alpha^2).$$

($\alpha \leftrightarrow |\mathcal{T}_0|^2$)

$$0 = \frac{d\ln|\mathcal{T}|^2_{\text{obs}}}{d\ln\mu} = \frac{2}{\alpha}\frac{d\alpha}{d\ln\mu} + 3\alpha + O(\alpha^2).$$

$$\rightarrow \frac{d\alpha}{d\ln\mu} = -\frac{3}{2}\alpha^2 + O(\alpha^3)$$

beta function

$$\rightarrow \alpha(\mu_2) = \frac{\alpha(\mu_1)}{1 + \frac{3}{2}\alpha(\mu_1)\ln\mu_2/\mu_1}$$

with b.c. that $\mu_2 = \mu_1 \rightarrow \alpha(\mu_2) = \alpha(\mu_1)$

Physically, if one determines α at scale μ_1
(typically ~~it's~~ we take $\mu = s$) (1)
the scale at which we do the experiment.

we know how it should be at another scale μ_2.

Physics becomes predictable!

1 Should read $\mu \sim \sqrt{s}$ (μ is a mass): then $\ln(s/\mu^2) = 0$ on p. 10 and $|\mathcal{T}_{\text{obs}}|^2 = |\mathcal{T}_0|^2_{\alpha(\sqrt{s})}[1 - \frac{\alpha}{3}\ln\delta^{-2}]$. p. 3's $\frac{3}{2}\ln(s/m^2)$ is just α^2 running from m to $\sqrt{s}$.

Here for beta function negative, the theory is more and more weakly coupled at higher and higher energy.

We call it asymptotic freedom (QCD is this kind).

[1] Should read: $SU(2)_L$ is asymptotically free, like QCD: $\beta_g = bg^3/16\pi^2$, $b = -\frac{22}{3} + 4 + \frac{1}{6} = -\frac{19}{6}$ (gauge bosons, three generations, Higgs). Only U(1) (QED, hypercharge) is IR free.

On the contrary, if beta is positive, the theory is more and more strongly coupled and we have to deal with it differently at high energy. It is infrared free, but UV-strongly coupled. ($U(1)_{EM}$ is such kind, as well $SU(2)_L$ weak) (1).

but for lower and lower energies, the theory becomes strongly coupled and our perturbative description fails. Luckily, we know the effective theory at low energy for QCD., that is mesons & baryons. Of particular relevance to us, protons & neutrons.

Remember that all $m(\mu)$, $\alpha(\mu)$ ~~etc~~ varies. So $m_{\overline{MS}}$ is a parameter of the theory but not physical.

[2] Should read: a fixed point is $\beta(\alpha_*) = 0$ with $m = 0$, since m/μ always runs: $d\ln(m/\mu)/d\ln\mu = \gamma_m - 1$. $\gamma_m(\alpha_*)$, $\gamma_\varphi(\alpha_*)$ need not vanish; they fix the scaling dimensions of φ and φ^2.

One has to be careful we discussing quantities.

What if $\gamma_m = 0$, $\beta_\alpha = 0$,? if they are strickly zero. (2) we are at fixed point of a theory. Lovely core of CFT (Conformal Theory

The discussion is reasonably physical but very complex. Now let's view it from a different angle:

RG

We have been dealing with two Lagrangians.

Renormalized (MS) $\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \, \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 + \frac{1}{3!} Z_g g \mu^{n\epsilon/2} \varphi^3 + Y\varphi$

and

Bare Lagrangian $\mathcal{L} = -\frac{1}{2} \partial^\mu \varphi_0 \, \partial_\mu \varphi_0 - \frac{1}{2} m_0^2 \varphi_0^2 + \frac{1}{3!} g_0 \varphi_0^3 + Y_0 \varphi_0$

By comparison, ~~if we~~ if

$$\varphi_0(x) = Z_\varphi^{1/2} \varphi(x)$$

$$m_0 = Z_\varphi^{-1/2} Z_m^{1/2} m$$

$$g_0 = Z_\varphi^{-3/2} Z_g \, g \mu^{n\epsilon/2}$$

$$Y_0 = Z_\varphi^{-1/2} Y.$$

These two would be equivalent, which means so long as my bare Lagrangian, which is physical if parameters are written properly would be fine and predictive.

Recall that in $\overline{MS}$ we just cancel the divergencies labelled by $1/\epsilon$, generally,

$$Z_\psi = 1 + \sum_{n=1}^{\infty} \frac{a_n(\alpha)}{\epsilon^n}$$

$$Z_m = 1 + \sum_{n=1}^{\infty} \frac{b_n(\alpha)}{\epsilon^n}$$

$$Z_g = 1 + \sum_{n=1}^{\infty} \frac{c_n(\alpha)}{\epsilon^n}, \quad \alpha = \frac{g^2}{(4\pi)^3}$$

Our calculations of $\Pi(k^2)$ and $\Gamma_3(k_1, k_2, k_3)$ in perturbative theory in the $\overline{MS}$ scheme, give us Taylor series in α for $a_n(\alpha)$, $b_n(\alpha)$ and $c_n(\alpha)$.

And we found

$$a_1 = -\frac{1}{6}\alpha + O(\alpha^2)$$

$$b_1 = -\alpha + O(\alpha^2)$$

$$c_1 = -\alpha + O(\alpha^2)$$

Here is the core physical condition. Bare parameters has to be independent from μ.

$$\alpha_0 \equiv g_0^2/(4\pi)^3 = Z_g^2 Z_\psi^{-3} \mu^{2\epsilon} \alpha.$$

$$\ln(Z_g^2 Z_\psi^{-3}) \equiv G(\alpha, \epsilon) \equiv \sum_{n=1}^{\infty} \frac{G_n(\alpha)}{\epsilon^n}$$

$$G_1(\alpha) = 2G_1(\alpha) - 3\alpha a_1(\alpha) = -\frac{3}{2}\alpha + O(\alpha^2).$$

Now we know

$$\ln \alpha_0 = G(\alpha, \epsilon) + \ln\alpha + \frac{\epsilon \ln \tilde{\mu}}{\epsilon \ln \mu + \epsilon C}.$$

$$0 = \frac{d\ln\alpha_0}{d\ln\mu} = \frac{\partial G(\alpha,\epsilon)}{\partial \alpha}\frac{d\alpha}{d\ln\mu} + \frac{1}{\alpha}\frac{d\alpha}{d\ln\mu} + \epsilon.$$

$$\Rightarrow \quad 0 = \left(1 + \frac{\alpha G_1'(\alpha)}{\epsilon} + \frac{\alpha G_2'(\alpha)}{\epsilon^2} + \dots\right)\frac{d\alpha}{d\ln\mu} + \epsilon\alpha$$

In the $G \to 0$ limit, α should move slowly

$$\frac{d\alpha}{d\ln\mu} = -\epsilon\alpha + \beta(\alpha) \quad (1)$$

1 Why $-\epsilon\alpha + \beta(\alpha)$? α is finite at $\epsilon \to 0$, so $d\alpha/d\ln\mu = \sum_{k\geq 0} f_k(\alpha)\epsilon^k$. In the line above, the top power $\epsilon^k f_k$ ($k \geq 2$) has nothing to cancel it: $f_k = 0$. Order ϵ: $f_1 = -\alpha$, the tree-level $\alpha_0 = \tilde{\mu}^\epsilon \alpha$. Order ϵ^0: $\beta = \alpha^2 G_1'$; order $1/\epsilon$: $G_2' = \alpha G_1'^2$.

~~$\beta(\alpha) = \alpha^2 G_1'(\alpha)$~~

they should order by order in $\frac{1}{\epsilon}$ cancel. hence

$$\beta(\alpha) = \alpha^2 G_1'(\alpha)$$

and $\quad G_2'(\alpha) = \alpha G_1'(\alpha)^2 \dots$

$$\beta(\alpha) = -\frac{3}{2}\alpha^2 + O(\alpha^3).$$

Similarly, one can derive various $d\ln$ [illegible]

$$d\ln \tilde{\Delta}_0(k^2)$$

$$\Rightarrow \left(\frac{\partial}{\partial \ln\mu} + \beta(\alpha)\frac{\partial}{\partial\alpha} + \gamma_m(\alpha)\, m\frac{\partial}{\partial m} + 2\gamma_\varphi(\alpha)\right)\tilde{\Delta}(k^2) = 0. \quad (1)$$

Callan-Symanzik-Equation.

1 $\gamma_\varphi \equiv \frac{1}{2} d\ln Z_\varphi / d\ln\mu = \frac{\alpha}{12}$, from $\tilde{\mathbf{\Delta}}_0 = Z_\varphi \tilde{\mathbf{\Delta}}$; $\gamma_m = \frac{1}{2} d\ln(Z_\varphi/Z_m)/d\ln\mu$. Checked: the operator annihilates the one-loop $\tilde{\mathbf{\Delta}}_{\overline{\text{MS}}}$. Payoff: R runs with $-2\gamma_\varphi$, so at $\mu^2 = s$, $\mathcal{T} \approx \mathcal{T}_0|_{\alpha(\sqrt{s})}[\alpha(\sqrt{s})/\alpha(m)]^{2/9}$; expanded, $-(\frac{3}{4}+\frac{1}{6})\alpha\ln(s/m^2)$: p. 1's large log, resummed.